Topic 13A: Lattice energy
1. be able to define lattice energy as the energy change when one mole of an ionic solid is formed from its gaseous ions
2. be able to define the terms: i enthalpy change of atomisation, $\Delta_{at}H$ ii electron affinity
3. be able to construct Born-Haber cycles and carry out related calculations
4. know that lattice energy provides a measure of ionic bond strength
5. understand that a comparison of the experimental lattice energy value (from a Born-Haber cycle) with the theoretical value (obtained from electrostatic theory) in a particular compound indicates the degree of covalent bonding
6. understand the meaning of polarisation as applied to ions
7. know that the polarising power of a cation depends on its radius and charge
8. know that the polarisability of an anion depends on its radius and charge
9. be able to define the terms 'enthalpy change of solution, $\Delta_{sol}H$', and 'enthalpy change of hydration, $\Delta_{hyd}H$'
10. be able to use energy cycles and energy level diagrams to carry out calculations involving enthalpy change of solution, enthalpy change of hydration and lattice energy
11. understand the effect of ionic charge and ionic radius on the values of: i lattice energy ii enthalpy change of hydration